IS42/45R86400D/16320D/32160D
IS42/45S86400D/16320D/32160D

16Mx32, 32Mx16, 64Mx8
512Mb SDRAM

PRELIMINARY INFORMATION
JUNE 2011

FEATURES

- Clock frequency: 200, 166, 143 MHz
- Fully synchronous; all signals referenced to a positive clock edge
- Internal bank for hiding row access/precharge
- Power supply: V_{DD}/V_{DDQ} = 2.3V-3.6V
  IS42/45SxxxxxD - V_{DD}/V_{DDQ} = 3.3V
  IS42/45RxxxxxD - V_{DD}/V_{DDQ} = 2.5
- LVTTL interface
- Programmable burst length – (1, 2, 4, 8, full page)
- Programmable burst sequence: Sequential/Interleave
- Auto Refresh (CBR)
- Self Refresh
- 8K refresh cycles every 64 ms
- Random column address every clock cycle
- Programmable $\overline{CAS}$ latency (2, 3 clocks)
- Burst read/write and burst read/single write operations capability
- Burst termination by burst stop and precharge command
- Packages:
  x8/x16: 54-pin TF-TSOP-II, 54-ball TF-BGA (x16 only)
  x32: 90-ball TF-BGA
- Temperature Range:
  Commercial (0°C to +70°C)
  Industrial (-40°C to +85°C)
  Automotive, A1 (-40°C to +85°C)
  Automotive, A2 (-40°C to +105°C)

DEVICE OVERVIEW

ISSI's 512Mb Synchronous DRAM achieves high-speed data transfer using pipeline architecture. All inputs and outputs signals refer to the rising edge of the clock input. The 512Mb SDRAM is organized as follows.

PACKAGE INFORMATION

IS42/45S32160D	IS42/45S16320D	IS42/45S86400D
IS42/45R32160D	**IS42/45R16320D**	**IS42/45R86400D**
4M x 32 x 4 banks	8M x 16 x 4 banks	16M x 8 x 4 banks
90-ball TF-BGA	54-pin TSOP-II 54-ball TF-BGA	54-pin TSOP-II

KEY TIMING PARAMETERS

Parameter	-5	-6	-7	Unit
Clk Cycle Time				
$\overline{CAS}$ Latency = 3	5	6	7	ns
$\overline{CAS}$ Latency = 2	10	10	7.5	ns
Clk Frequency				
$\overline{CAS}$ Latency = 3	200	167	143	Mhz
$\overline{CAS}$ Latency = 2	100	100	133	Mhz
Access Time from Clock				
$\overline{CAS}$ Latency = 3	5.0	5.4	5.4	ns
$\overline{CAS}$ Latency = 2	6	6	5.4	ns

ADDRESS TABLE

Parameter	16M x 32	32M x 16	64M x 8
Configuration	4M x 32 x 4 banks	8M x 16 x 4 banks	16M x 8 x 4 banks
Bank Address Pins	BA0, BA1	BA0, BA1	BA0, BA1
Autoprecharge Pins	A10/AP	A10/AP	A10/AP
Row Address	8K(A0 – A12)	8K(A0 – A12)	8K(A0 – A12)
Column Address	512(A0 – A8)	1K(A0 – A9)	2K(A0 – A9, A11)
Refresh Count			
Com./Ind./A1	8K / 64ms	8K / 64ms	8K / 64ms
A2	8K / 16ms	8K / 16ms	8K / 16ms

PIN FUNCTIONS

Symbol	Type	Function (In Detail)
A0-A12	Input Pin	Address Inputs: A0-A12 are sampled during the ACTIVE command (row-address A0-A12) and READ/WRITE command (column address A0-A9, A11 (x8); A0-A9 (x16); A0-A8 (x32); with A10 defining auto precharge) to select one location out of the memory array in the respective bank. A10 is sampled during a PRECHARGE command to determine if all banks are to be precharged (A10 HIGH) or bank selected by BA0, BA1 (LOW). The address inputs also provide the op-code during a LOAD MODE REGISTER command.
BA0, BA1	Input Pin	Bank Select Address: BA0 and BA1 defines which bank the ACTIVE, READ, WRITE or PRECHARGE command is being applied.
$\overline{CAS}$	Input Pin	$\overline{CAS}$, in conjunction with the $\overline{RAS}$ and $\overline{WE}$, forms the device command. See the "Command Truth Table" for details on device commands.
CKE	Input Pin	The CKE input determines whether the CLK input is enabled. The next rising edge of the CLK signal will be valid when is CKE HIGH and invalid when LOW. When CKE is LOW, the device will be in either power-down mode, clock suspend mode, or self refresh mode. CKE is an asynchronous input.
CLK	Input Pin	CLK is the master clock input for this device. Except for CKE, all inputs to this device are acquired in synchronization with the rising edge of this pin.
$\overline{CS}$	Input Pin	The $\overline{CS}$ input determines whether command input is enabled within the device. Command input is enabled when $\overline{CS}$ is LOW, and disabled with $\overline{CS}$ is HIGH. The device remains in the previous state when $\overline{CS}$ is HIGH.
DQM: x8 DQML, DQMH: x16 DQM0-DQM3: x32	Input Pin	DQx pins control the bytes of the I/O buffers. For example with x16, in read mode, DQML and DQMH control the output buffer. When DQML or DQMH is LOW, the corresponding buffer byte is enabled, and when HIGH, disabled. The outputs go to the HIGH impedance state when DQML/DQMH is HIGH. This function corresponds to $\overline{OE}$ in conventional DRAMs. In write mode, DQML and DQMH control the input buffer. When DQML or DQMH is LOW, the corresponding buffer byte is enabled, and data can be written to the device. When DQML or DQMH is HIGH, input data is masked and cannot be written to the device.
DQ_0-DQ_7: x8 DQ_0-DQ_{15}: x16 DQ_0-DQ_{31}: x32	Input/Output	Data on the Data Bus is latched on DQ pins during Write commands, and buffered for output after Read commands.
$\overline{RAS}$	Input Pin	$\overline{RAS}$, in conjunction with $\overline{CAS}$ and $\overline{WE}$, forms the device command. See the "Command Truth Table" item for details on device commands.
$\overline{WE}$	Input Pin	$\overline{WE}$, in conjunction with $\overline{RAS}$ and $\overline{CAS}$, forms the device command. See the "Command Truth Table" item for details on device commands.
V_{DDQ}	Power Supply Pin	V_{DDQ} is the output buffer power supply.
V_{DD}	Power Supply Pin	V_{DD} is the device internal power supply.
V_{SSQ}	Power Supply Pin	V_{SSQ} is the output buffer ground.
V_{SS}	Power Supply Pin	V_{SS} is the device internal ground.

AC ELECTRICAL CHARACTERISTICS (1,2,3)

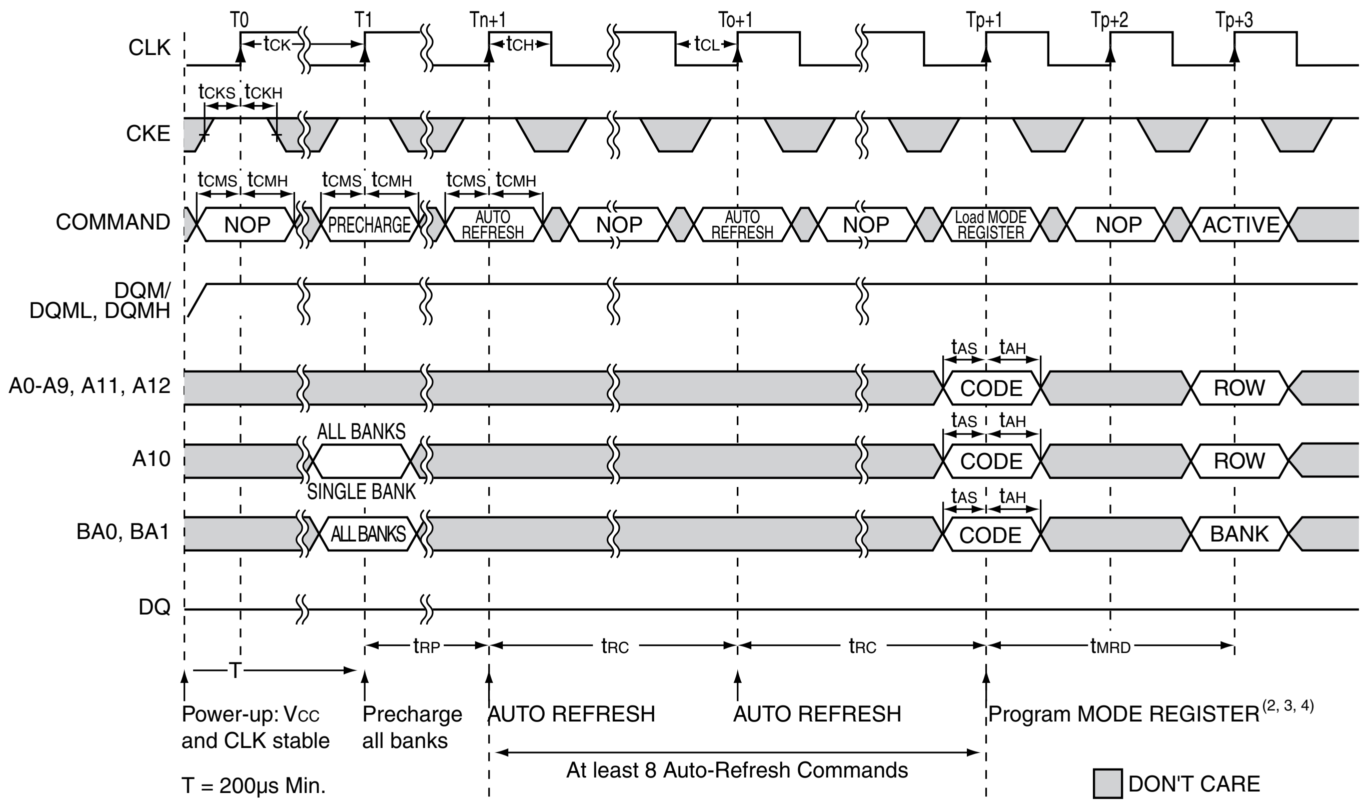
			-5		-6		-7		
Symbol	Parameter		Min.	Max.	Min.	Max.	Min.	Max.	Units
t_{CK3}	Clock Cycle Time	$\overline{CAS}$ Latency = 3	5	—	6	—	7	—	ns
t_{CK2}		$\overline{CAS}$ Latency = 2	10	—	10	—	7.5	—	ns
t_{AC3}	Access Time From CLK	$\overline{CAS}$ Latency = 3	—	5.0	—	5.4	—	5.4	ns
t_{AC2}		$\overline{CAS}$ Latency = 2	—	6	—	6	—	5.4	ns
t_{CH}	CLK HIGH Level Width		2	—	2.5	—	2.5	—	ns
t_{CL}	CLK LOW Level Width		2	—	2.5	—	2.5	—	ns
t_{OH3}	Output Data Hold Time	$\overline{CAS}$ Latency = 3	2.5	—	2.7	—	2.7	—	ns
t_{OH2}		$\overline{CAS}$ Latency = 2	2.5	—	2.7	—	2.7	—	ns
t_{LZ}	Output LOW Impedance Time		0	—	0	—	0	—	ns
t_{HZ3}	Output HIGH Impedance Time		—	5	—	5.4	—	5.4	ns
t_{HZ2}			—	6	—	6	—	5.4	ns
t_{DS}	Input Data Setup Time(2)		1.8	—	1.5	—	1.5	—	ns
t_{DH}	Input Data Hold Time(2)		0.8	—	0.8	—	0.8	—	ns
t_{AS}	Address Setup Time(2)		1.5	—	1.5	—	1.5	—	ns
t_{AH}	Address Hold Time(2)		0.8	—	0.8	—	0.8	—	ns
t_{CKS}	CKE Setup Time(2)		1.5	—	1.5	—	1.5	—	ns
t_{CKH}	CKE Hold Time(2)		0.8	—	0.8	—	0.8	—	ns
t_{CMS}	Command Setup Time ($\overline{CS}$, $\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$, DQM)(2)		1.5	—	1.5	—	1.5	—	ns
t_{CMH}	Command Hold Time ($\overline{CS}$, $\overline{RAS}$, $\overline{CAS}$, $\overline{WE}$, DQM)(2)		0.8	—	0.8	—	0.8	—	ns
t_{RC}	Command Period (REF to REF / ACT to ACT)		55	—	60	—	60	—	ns
t_{RAS}	Command Period (ACT to PRE)		38	100K	42	100K	37	100K	ns
t_{RP}	Command Period (PRE to ACT)		15	—	18	—	15	—	ns
t_{RCD}	Active Command To Read / Write Command Delay Time		15	—	18	—	15	—	ns
t_{RRD}	Command Period (ACT [0] to ACT[1])		10	—	12	—	14	—	ns
t_{DPL}	Input Data To Precharge Command Delay time		10	—	12	—	14	—	ns
t_{DAL}	Input Data To Active / Refresh Command Delay time (During Auto-Precharge)		25	—	30	—	29	—	ns
t_{MRD}	Mode Register Program Time		10	—	12	—	14	—	ns
t_{DDE}	Power Down Exit Setup Time		5	—	6	—	7	—	ns
t_{XSR}	Self-Refresh Exit Time		60	—	70	—	67	—	ns
t_{T}	Transition Time		0.3	1.2	0.3	1.2	0.3	1.2	ns
t_{REF}	Refresh Cycle Time (8192)	$T_A \leq 70^oC$ Com, Ind, A1, A2	—	64	—	64	—	64	ms
		$T_A \leq 85^oC$ Ind, A1, A2	—	64	—	64	—	64	ms
		$T_A > 85^oC$ A2	—	—	—	—	—	16	ms

Notes:
1. The power-on sequence must be executed before starting memory operation.
2. Measured with t_T = 1 ns. If clock rising time is longer than 1ns, (t_T /2 - 0.5) ns should be added to the parameter.
3. The reference level is 1.4V when measuring input signal timing. Rise and fall times are measured between V_{IH}(min.) and V_{IL} (max).

OPERATING FREQUENCY / LATENCY RELATIONSHIPS

SYMBOL	PARAMETER		-5	-6	-7	-7	UNITS
—	Clock Cycle Time		5	6	7	7.5	ns
—	Operating Frequency		200	167	143	133	MHz
t_{CAC}	$\overline{CAS}$ Latency		3	3	3	2	cycle
t_{RCD}	Active Command To Read/Write Command Delay Time		3	3	3	2	cycle
t_{RAC}	$\overline{RAS}$ Latency (t_{RCD} + t_{CAC})	$\overline{CAS}$ Latency = 3 $\overline{CAS}$ Latency = 2	6 —	6 —	6 —	— 4	cycle
t_{RC}	Command Period (REF to REF / ACT to ACT)		10	10	9	8	cycle
t_{RAS}	Command Period (ACT to PRE)		7	7	6	5	cycle
t_{RP}	Command Period (PRE to ACT)		3	3	3	2	cycle
t_{RRD}	Command Period (ACT[0] to ACT [1])		2	2	2	2	cycle
t_{CCD}	Column Command Delay Time (READ, READA, WRIT, WRITA)		1	1	1	1	cycle
t_{DPL}	Input Data To Precharge Command Delay Time		2	2	2	2	cycle
t_{DAL}	Input Data To Active/Refresh Command Delay Time (During Auto-Precharge)		5	5	5	4	cycle
t_{RBD}	Burst Stop Command To Output in HIGH-Z Delay Time (Read)	$\overline{CAS}$ Latency = 3 $\overline{CAS}$ Latency = 2	3 —	3 —	3 —	— 2	cycle
t_{WBD}	Burst Stop Command To Input in Invalid Delay Time (Write)		0	0	0	0	cycle
t_{RQL}	Precharge Command To Output in HIGH-Z Delay Time (Read)	$\overline{CAS}$ Latency = 3 $\overline{CAS}$ Latency = 2	3 —	3 —	3 —	— 2	cycle
t_{WDL}	Precharge Command To Input in Invalid Delay Time (Write)		0	0	0	0	cycle
t_{PQL}	Last Output To Auto-Precharge Start Time (Read)	$\overline{CAS}$ Latency = 3 $\overline{CAS}$ Latency = 2	-2 —	-2 —	–2 —	— -1	cycle
t_{QMD}	DQM To Output Delay Time (Read)		2	2	2	2	cycle
t_{DMD}	DQM To Input Delay Time (Write)		0	0	0	0	cycle
t_{MRD}	Mode Register Set To Command Delay Time		2	2	2	2	cycle

INITIALIZE AND LOAD MODE REGISTER[1]



Notes:

1. If $\overline{CS}$ is High at clock High time, all commands applied are NOP.
2. The Mode register may be loaded prior to the Auto-Refresh cycles if desired.
3. JEDEC and PC100 specify three clocks.
4. Outputs are guaranteed High-Z after the command is issued.

REGISTER DEFINITION

Mode Register

The mode register is used to define the specific mode of operation of the SDRAM. This definition includes the selection of a burst length, a burst type, a CAS latency, an operating mode and a write burst mode, as shown in MODE REGISTER DEFINITION.

The mode register is programmed via the LOAD MODE REGISTER command and will retain the stored information until it is programmed again or the device loses power.

Mode register bits M0-M2 specify the burst length, M3 specifies the type of burst (sequential or interleaved), M4- M6 specify the CAS latency, M7 and M8 specify the operating mode, M9 specifies the WRITE burst mode, and M10, M11, and M12 are reserved for future use.

The mode register must be loaded when all banks are idle, and the controller must wait the specified time before initiating the subsequent operation. Violating either of these requirements will result in unspecified operation.

MODE REGISTER DEFINITION

BA1	BA0	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0	Address Bus (Ax)
Reserved(1)					Write Burst Mode	Operating Mode		Latency Mode			Burst Type	Burst Length			Mode Register (Mx)

Burst Length

M2	M1	M0	M3=0	M3=1
0	0	0	1	1
0	0	1	2	2
0	1	0	4	4
0	1	1	8	8
1	0	0	Reserved	Reserved
1	0	1	Reserved	Reserved
1	1	0	Reserved	Reserved
1	1	1	Full Page	Reserved

Burst Type

M3	Type
0	Sequential
1	Interleaved

Latency Mode

M6	M5	M4	CAS Latency
0	0	0	Reserved
0	0	1	Reserved
0	1	0	2
0	1	1	3
1	0	0	Reserved
1	0	1	Reserved
1	1	0	Reserved
1	1	1	Reserved

Operating Mode

M8	M7	M6-M0	Mode
0	0	Defined	Standard Operation
—	—	—	All Other States Reserved

Write Burst Mode

M9	Mode
0	Programmed Burst Length
1	Single Location Access

1. To ensure compatibility with future devices, should program BA1, BA0, A12, A11, A10 = "0"